

What Does MMLU Actually Measure?

A Psychometric Audit of Difficulty Structure in Aggregate Benchmark Scores

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Abstract

Although MMLU is widely adopted as a benchmark for calibrating general AI capabilities, we psychometrically demonstrate that its aggregate score primarily evaluates a model’s factual retrieval capacity rather than its reasoning ability. By calibrating item difficulty for 1,000 open-weights language models over 14,042 MMLU test items using Item Response Theory, we show that evaluating both abilities via a single test is inherently flawed. Difficulty is then regressed on a deterministic, text-extractable framework of structural complexity. Applying a joint Wald test with subject-clustered covariances demonstrates that the MMLU conflates fundamentally separable constructs. The mapping from structural complexity to difficulty is not invariant across the benchmark’s STEM and non-STEM partitions. This finding has practical consequences. Aggregate leaderboard ranks track non-STEM accuracy more closely than STEM accuracy, so selecting a Top-50 model on the aggregate for a reasoning-intensive deployment displaces roughly 22% of the STEM-appropriate choices. Furthermore, when controlling for the multiple-choice guessing floor natively inside the response model, we find that higher-ability models continue to degrade more steeply under increased reasoning depth. The MMLU aggregate therefore weights retrieval capacity and reasoning stability unequally, inadvertently favoring models optimized for retrieval. We release our deterministic framework as a reproducible auditing instrument and recommend disaggregated reporting.

evaluates large language models (LLMs) on 14,042 multiple-choice items across 57 subjects and reports a single aggregate accuracy score, which practitioners widely read as a unified measure of general capability. Whether one score psychometrically captures a coherent construct is an old question: the coherence of a single ability score has been contested since [Spearman \(1904\)](#) introduced a general intelligence factor g , which [Thurstone \(1938\)](#) and [Carroll \(1993\)](#) showed dissolves into separable primary abilities under factor analysis. ACT-R draws the distinction that matters here, separating declarative knowledge, the retrieval of facts, from procedural knowledge, the execution of multi-step logical operations ([Anderson, 1983](#); [Tulving, 1972](#)), and we operationalize it directly: Zipf Rarity and Entity Density capture a prompt’s declarative retrieval demands, while the Weighted Symbolic Complexity Graph (WSCG) captures its sequential inferential load. If MMLU items draw unequally on these processes, aggregating them into a single metric risks conflating separable abilities.

There is also empirical reason to expect MMLU in particular to be retrieval-weighted. [Wang et al. \(2024\)](#) documented that Chain-of-Thought prompting fails to improve, and sometimes degrades, MMLU accuracy while yielding large gains on their successor MMLU-Pro, concluding that MMLU is “mostly knowledge-driven without requiring too much reasoning.” That diagnosis was nonetheless reached heuristically: MMLU-Pro filtered on item difficulty (b_i) rather than on discrimination (a_i), which restricts the measurement range instead of deleting items that fail to separate models, and no psychometric criterion was specified for differentiating a “knowledge-driven” item from a “reasoning-driven” one.

To investigate exactly what MMLU psychometrically measures, we conduct an explanatory item-response audit of its difficulty structure. Building

1 Introduction

The Massive Multitask Language Understanding (MMLU) benchmark ([Hendrycks et al., 2021a](#))

on the foundational application of Item Response Theory (IRT) to NLP (Lalor et al., 2019), we calibrate item difficulty and model ability with a Two-Parameter Logistic (2PL) model fitted to roughly 14 million responses from 1,000 open-weights models on the Open LLM Leaderboard (Beeching et al., 2023), and regress the estimated difficulties on a deterministic, text-extractable framework of five dimensions and nine indicators. We then test whether that mapping is structurally invariant across domains, adopting the benchmark designers’ *a priori* taxonomy rather than a post hoc split, on the hypothesis that STEM items generally demand higher procedural reasoning while non-STEM items rely more heavily on factual retrieval. The dichotomy is imperfect, since College Biology (STEM) requires substantial recall whereas Formal Logic (non-STEM) requires inferential chaining, but it is a predefined, unbiased boundary against which homogeneity can fail.

Our analysis yields three findings. First, MMLU’s two designer-defined partitions do not measure a single interchangeable ability: a confirmatory two-dimensional IRT model, in which each item loads only on the dimension its designer-assigned subject specifies, puts the latent correlation between STEM and non-STEM ability at $\rho = 0.965$, strictly below unity with no posterior mass above 0.99, predicts held-out responses better than the unidimensional model, and is corroborated by a disattenuated subscale correlation of 0.977 at matched test length. Second, the mapping from text structure to item difficulty is not the same function in the two partitions, and the one contrast surviving subject-clustered covariances, a wild cluster bootstrap and Holm correction over the family of nine is Entity Density, which acts as an active constraint inside a STEM item and as a retrieval cue outside one. Third, aggregate rank tracks non-STEM accuracy more closely than STEM accuracy, even after the partitions are equalized for test length. We release the framework, the features and the IRT estimates, so that this audit can be repeated on any benchmark whose items are text.

2 Related Work

Aggregate benchmark scores masking construct conflation is well documented. Bender et al. (2021) argue that knowledge-retrieval performance is no evidence of inferential capacity, McCoy et al. (2019) showed BERT-family models relying on sur-

face heuristics rather than logical inference on NLI benchmarks, and high GLUE scores have been catalogued beside near-random performance on reasoning probes (Rogers et al., 2020); the strongest statement is Bowman and Dahl’s (2021), that benchmarks framed as measures of general ability rarely possess the construct validity such framing presupposes (see also Raji et al., 2021). We present psychometric evidence corroborating those critiques. Lalor et al. (2019) pioneered IRT for NLP benchmarks, Rodriguez et al. (2021) built leaderboards on that observation and Polo et al. (2024) extended the line with tinyBenchmarks, but all three assume unidimensionality, i.e. a single latent ability θ_j per model, and hence cannot detect whether that dimension conflates separable constructs contributing non-uniformly across domains. Exploratory multidimensional IRT (Reckase, 2009) recovers its factors from the response matrix and interprets them post hoc, which makes the dimensions a product of the analysis rather than a hypothesis the data can reject; we therefore fit a confirmatory model whose loading pattern is fixed by MMLU’s own designers (Appendix C).

What we are and are not testing. Our procedure is a multi-group test of an explanatory item-difficulty model, i.e. a Linear Logistic Test Model (Fischer, 1973) extended with an error term and estimated separately in two groups of items (De Boeck and Wilson, 2004). It is the item-side analogue of measurement invariance rather than invariance itself (Meredith, 1993; Vandenberg and Lance, 2000): the respondent population is fixed, the groups are groups of items, and the null holds constant a regression of item parameters on item covariates. The respondent-side test is separately available to us with model families in place of examinee groups, and Appendix D reports Mantel-Haenszel differential item functioning across architecture families (Holland and Thayer, 1988; Dorans and Holland, 1993). The relevant validity is construct representation rather than nomothetic span (Embretson, 1983).

Predicting item difficulty from text has precedent in educational measurement (Embretson and Wetzel, 1987) and in NLP (Vania et al., 2021), while domain-targeted evaluations such as ARC (Clark et al., 2018) and AGIEval (Zhong et al., 2024) instead construct item pools isolating particular reasoning types. Our syntactic indicators rest on Dependency Locality Theory (Gibson, 1998; Dem-

berg and Keller, 2008), operationalized as Mean Dependency Distance (Liu, 2008): since written English minimizes dependency length wherever grammatically possible (Temperley, 2007; Gildea and Jaeger, 2015), unusually long dependencies mark a departure from the baseline (cf. Levy, 2008). Finally, MMLU rankings are sensitive to semantically invariant perturbations such as prompt templates and option ordering (Alzahrani et al., 2024; Pezeshkpour and Hruschka, 2024; Zheng et al., 2024; Mizrahi et al., 2024), so Section 5.2 measures a noise floor inside our own population, and Appendix D refits the structural model against the 6.49% label error rate of Gema et al. (2025) and the contamination documented by Sainz et al. (2023), Golchin and Surdeanu (2024) and Deng et al. (2024).

3 Measurement Framework and Methodology

3.1 Data and Latent Parameter Calibration

We use the complete MMLU test set ($I = 14,042$ items across 57 subjects) and a model population of $J = 1,000$ open-weights models from the Hugging Face Open LLM Leaderboard (Beeching et al., 2023), spanning the Deepseek, Falcon, Llama, Mistral, Pythia, Qwen, Solar, Yi and Mixture-of-Experts families, with parameter counts from 3.2 million to over 180 billion, so that the binary response matrix $\mathbf{Y} \in \{0, 1\}^{J \times I}$ yields approximately 14 million observations. That population is a leaderboard rather than a designed sample and is substantially redundant, reducing to 869 distinct checkpoints and 50 base-family-and-size combinations; redundancy is harmless for item parameters estimated from a thousand models but inflates any quantity counted over models, so every model-level statistic below is reported on the deduplicated population as well.

Item difficulty is estimated using the Two-Parameter Logistic model (Birnbau, 1968; Lord and Novick, 1968), $P(Y_{ji} = 1 \mid \theta_j, a_i, b_i) = [1 + \exp(-a_i(\theta_j - b_i))]^{-1}$, where b_i is the difficulty of item i and $a_i > 0$ its discrimination, with priors $\theta_j \sim \mathcal{N}(0, 1)$, $b_i \sim \mathcal{N}(0, 1)$ and $a_i \sim \text{HalfNormal}(1.0)$; with 1,000 models per item the likelihood dominates the prior. We estimate by Stochastic Variational Inference in NumPyro and JAX rather than Joint Maximum Likelihood, since JML suffers from the incidental parameters problem (Neyman and Scott, 1948), the nuisance pa-

rameters $\{\theta_j\}$ increasing with J so that item estimates remain inconsistent, whereas SVI marginalizes over the ability distribution. Calibrated difficulty spans $b_i \in [-5.00, 7.12]$ ($\sigma = 2.24$) and ability spans $\theta_j \in [-2.02, 3.14]$ ($\sigma = 1.03$). Calibrating two independent random cohorts of 500 models separately, the two difficulty vectors correlate at $r_{hh} = 0.9921$ across all 14,042 items, a full-test reliability of approximately 0.9960 under Spearman–Brown. The estimates are therefore mostly free of incidental measurement error from model sampling, and downstream disattenuation corrections are not needed.

3.2 A Priori Domain Grouping

To prevent post hoc selection bias on observed response patterns or text features, we fix the subject partition before any path model is estimated, adopting the official taxonomy of MMLU’s designers (Hendrycks et al., 2021a), which gives $I_{\text{STEM}} = 3,153$ items across 19 subjects against $I_{\text{non-STEM}} = 10,889$ items across 38 subjects. It is well established in educational psychology (Newcombe et al., 2009) that mathematics and science learning recruit partly distinct cognitive processes from verbal and social reasoning. That partition is a union of whole subjects, and because MMLU items were written in subject blocks, items within a subject share topic, source, authoring and format; they are therefore not independent observations, the effective sample size is closer to 57 than to 14,042, and Section 4.3 accordingly treats the subject, not the item, as the unit of clustering and of resampling.

3.3 The Five-Dimension, Nine-Indicator Framework

We operationalize item complexity as five separable dimensions extracted deterministically from the prompt text, because two items can share identical word counts and syntactic structures while differing in logical nesting, lexical abstractness or factual obscurity. Table 1 gives each indicator’s definition and its cognitive or linguistic grounding; two construction details do not reduce to a table entry.

The Weighted Symbolic Complexity Graph. Dimension 1 measures how much sequential multi-step reasoning an item requires, since retrieving two facts is a different computational demand from chaining them through a nested conditional and no length or readability measure separates the two.

Indicator	Definition
<i>Dimension 1 — topological reasoning load</i> ; procedural knowledge in ACT-R (Anderson, 1983)	
WSCG Depth (W_{scg})	Weighted longest path $\max_{p \in \mathcal{P}} \sum_{v \in p} w(v)$ through the symbolic complexity DAG
WSCG Nodes (N_{scg})	Operational nodes in that DAG, i.e. distinct reasoning operations or leaves
<i>Dimension 2 — syntactic complexity</i> ; Dependency Locality Theory (Gibson, 1998)	
Syntactic MDD (D_{mdd})	$\frac{1}{n_i - 1} \sum_{t=2}^{n_i} t - h(t) $ over the Stanza parse, with $h(t)$ the head index of token t (Liu, 2008)
Syntactic Depth (H_{syn})	Longest leaf-to-root path in the dependency tree
<i>Dimension 3 — epistemological specificity</i> ; declarative knowledge in ACT-R (Anderson, 1983)	
Entity Density (D_{ner})	$ E_{ner} /N_{tokens}$, named-entity tokens from SpaCy NER as a fraction of all tokens
Zipf Rarity (R_{zipf})	$\frac{1}{ V_z } \sum_{w \in V_z} (8.0 - Z(w))$ over content words, Z the wordfreq Zipf frequency (Speer, 2022)
<i>Dimension 4 — semantic complexity</i> ; syntax/semantics double dissociation (Berndt et al., 1996)	
Concreteness (C_{bry})	Mean of the Brysbaert et al. (2014) 1–5 norms over covered content words, higher denoting more concrete
AMR Depth (D_{amr})	Longest path through the BART-AMR semantic DAG (Banarescu et al., 2013), i.e. nesting with surface syntax stripped
<i>Dimension 5 — adversarial logic structure</i> ; non-monotonic inference under negation	
Adversarial Negation ($\tilde{S}_{neg}$)	OLS residual, on token count, of raw negation scope counted over the dependency scope of a 59-term trigger lexicon

Table 1: Five dimensions and nine deterministic indicators of structural complexity, computed from the prompt stem.

For each prompt we construct a node-weighted directed acyclic graph $G_w = (V, E, w)$ over the Stanza parse, restricting V to operational tokens identified through a cognitive lexicon of 13 lexical categories plus a symbolic-operator class. Each vertex carries an intrinsic weight $w(v) \in [0.0, 5.0]$ on eight active tiers matching its theoretical inferential friction: boolean conjunctions take $w = 1.0$, causative conditionals $w = 4.0$ and L^AT_EX integrals $w = 5.0$. Edges are projected from grammatical dependency arcs and supplemented by state-tracking edges whenever an entity referenced in sentence i reappears in sentence $j > i$; because standard DAG path algorithms evaluate edge weights rather than node weights, WSCG Depth maps each target node’s weight onto its incoming edge before solving for the longest path. The weight assignments are heuristically motivated, a vulnerability we test directly: perturbing the tiers by ± 1.0 over 100 Monte Carlo draws, one tier at a time, leaves all three primary conclusions intact across all 123 schemes tested (Appendix A). The highest tier ($w = 5.0$, for integral and nabla operators) is inert here because MMLU mathematical content is overwhelmingly ASCII, so Dimension 1 is blind to a component of STEM difficulty, a scope condition we return to in Section 6.2.

Sensitivity to prompt length and to the answer options. Because nouns occupy the lowest active weight tier, one might object that topological reasoning load is merely correlated with prompt length. This holds for WSCG Nodes, which correlates with token count at $r = 0.985$, but

fails for WSCG Depth ($r = 0.580$), and replacing both Dimension 1 indicators with their OLS residuals on token count preserves the stratified fits ($W(9) = 134.79$), as does clamping the baseline 0.5 tier to zero. Separately, all nine indicators compute complexity from the question stem alone; eight option-set covariates yield complementary predictive power without eliminating the cross-domain contrast (Appendix B), and we restrict the primary analysis to the stem because its dimensions isolate theoretically grounded cognitive demand.

4 Results

4.1 Global Path Analysis and Baselines

We begin with a pooled OLS regression across all 14,042 items, regressing IRT difficulty on the nine-indicator framework as $b_i = \beta_0 + \sum_{k=1}^9 \beta_k X_{ki} + \epsilon_i$, with HC3 heteroskedasticity-consistent standard errors (MacKinnon and White, 1985); Section 4.3 replaces these with subject-clustered covariances for inferences that carry weight. Eight predictors reach significance and the omnibus test is extreme, yet the model explains 3.57% of global variance, which multicollinearity does not account for since variance inflation factors range from 1.025 to 2.379. Three considerations make a figure of this size the expected result rather than a failure. Educational measurement obtains R^2 in the 0.10–0.30 band from cognitive text features only on *homogeneous* pools (Embretson and Wetzel, 1987), whereas MMLU is 57 of them; the framework isolates only the *structural* component of diffi-

Predictor set	Global	STEM	n-STEM
Word count	0.0059	0.0123	0.0133
+ Flesch-Kincaid	0.0086	0.0289	0.0143
Nine indicators	0.0331	0.0732	0.0361
Indicators + surface	0.0383	0.0844	0.0360
Option-set indicators	0.0684	0.0825	0.0559
Indicators + options	0.0808	0.1196	0.0713
Subject identity only	0.1354	0.1246	0.1197
Encoder ceiling	0.1269	0.1330	0.1113
+ subject identity	0.1596	0.1584	0.1431

Table 2: Cross-validated R^2 (5-fold mean). The ceiling is a ridge regression on a frozen MiniLM encoder (stem + options).

culty, the remainder being content-specific; and the pooled coefficients are dominated by the majority regime, since 77.55% of items are non-STEM and a signal concentrated in STEM is diluted to near-invisibility.

The relevant question is what that figure should be compared against, and Table 2 answers it in two ways. Sheer length predicts difficulty weakly and a readability index adds a little more, chiefly in STEM; against that baseline the framework captures $3.8\times$ the cross-validated variance globally and $2.5\times$ in STEM, significantly in every partition (globally $F(9, 14029) = 51.15$, $p = 6.4 \times 10^{-92}$). The second comparison is a dense sentence encoder reading the whole item, which estimates the ceiling on text-predictable difficulty. That ceiling is itself low, at 0.1269 pooled and 0.1330 in STEM, so most MMLU item difficulty is not recoverable from item text by any method; measured against it, the nine indicators recover 56.5% of text-predictable difficulty in STEM and 32.6% outside it, and the option-set indicators take the STEM figure to 89.9%. Subject identity alone reaches 0.1354 pooled, so much of what an encoder knows about an item is which subject it came from, whereas the nine indicators carry no subject label; the informative comparison is therefore the framework against the ceiling.

4.2 Domain-Stratified Bifurcation

To recover the signal suppressed by pooling, we estimate separate OLS regressions on each partition (Table 3). Structural complexity explains roughly twice the cross-validated difficulty variance in STEM (0.0732) as outside it (0.0361), a ratio robust to the outcome’s scale, since in logit-squared units the indicators account for $2.46\times$ more variance in STEM (bootstrap CI [1.80, 3.36]). However, because MMLU is clustered by subject,

arbitrary 19-subject groupings routinely produce R^2 gaps of this size ($p = 0.08$; Appendix A). We therefore rest our structural claim on slope contrasts rather than on fit magnitude, having verified that standardizing within partitions leaves the identical five contrasts surviving Holm correction.

Entity Density (D_{ner}) shows the most robust divergence, surviving all corrections ($t_{\text{CR2}} = 3.41$, $p_{\text{Holm}} = 0.011$; wild cluster bootstrap $p = 0.014$). In STEM, named entities act as operational constraints compounding the reasoning chain ($\beta = 0.140$), whereas outside STEM the coefficient is indistinguishable from zero ($b = 0.0002$), entities there supplying associative context that facilitates recall rather than computational load; over their shared range the STEM relationship rises monotonically while the non-STEM fit traces a non-monotone curve (Appendix B). Syntactic MDD (D_{mdd}) similarly carries a strong positive coefficient in STEM ($b = 0.3694$) and near-zero outside it ($b = 0.0146$), clearing clustered inference ($t_{\text{CR2}} = 2.68$, bootstrap $p = 0.015$) though not Holm correction, so Dependency Locality Theory predictions regarding processing cost hold for procedural computation but not for factual recall. Lexical Concreteness (C_{bry}) reverses sign across the boundary ($Z_{\text{HC3}} = -4.94$), but the reversal does not survive subject clustering ($t_{\text{CR2}} = -1.93$) and a restricted cubic spline reveals the non-STEM relationship to be significantly non-linear (Appendix B). Four indicators do not differ under any estimator, and across the full set five contrasts clear Holm correction under HC3, two under CR1 and one under the bootstrap, so the bifurcation is specific instead of global.

Individual subject-level fits are descriptive only, since nine predictors on 100-item pools give an 11:1 observation-to-parameter ratio and severe in-sample optimism, and partition-level effect sizes are small (Cohen’s $q = 0.0918$, CI [0.052, 0.136], Cohen, 1988). The supportable claim is therefore not that domains differ in *how much* variance structure explains, but in *which* structural features do the explaining.

4.3 Testing Structural Homogeneity

We now test whether the coefficient differences in Table 3 constitute a significant violation of structural homogeneity, regressing difficulty on the nine indicators, a STEM dummy and their interactions, where homogeneity is the null that all nine in-

Complexity Indicator	Pooled	STEM		non-STEM		Contrast		
	b	b	β	b	β	Z_{HC3}	t_{CR2}	p_{boot}
WSCG Depth (W_{scg})	0.0696***	0.0602	0.073	0.0620	0.115	-0.08	-0.05	1.00
WSCG Nodes (N_{scg})	-0.0023*	0.0189	0.088	-0.0007	-0.010	2.93	1.76	0.46
Syntactic MDD (D_{mdd})	0.1654***	0.3694	0.102	0.0146	0.004	4.05	2.68	0.12
Syntactic Depth (H_{syn})	0.0027	-0.0584	-0.037	0.0130	0.012	-1.81	-1.28	0.84
Zipf Rarity (R_{zipf})	0.0971*	0.2523	0.047	-0.0521	-0.010	2.90	1.85	0.46
Entity Density (D_{ner})	0.0108***	0.0356	0.140	0.0002	0.001	7.03	3.41	0.014 [†]
Concreteness (C_{bry})	0.4127***	-0.1642	-0.021	0.5883	0.078	-4.94	-1.93	0.46
AMR Depth (D_{amr})	0.0516***	0.0523	0.035	0.0608	0.055	-0.28	-0.14	1.00
Adversarial Negation ($\tilde{S}_{\text{neg}}$)	-0.0232***	-0.0343	-0.033	-0.0177	-0.038	-0.84	-0.82	1.00

Table 3: OLS path coefficients for item difficulty (b : unstandardized; β : within-partition standardized). Contrasts test cross-domain homogeneity under progressively stricter assumptions: independent items (Z_{HC3}), 57-subject cluster-robust covariances (t_{CR2}), and a 9,999-replicate wild cluster bootstrap with Holm correction (p_{boot} ; [†] marks the sole survivor). *** $p < 0.001$, * $p < 0.05$.

teraction terms are zero. Assuming item independence, the HC3 Wald test decisively rejects it ($W(9) = 138.36$, $p = 2.3 \times 10^{-25}$); we exclude the intercept interaction ($W(10) = 375.76$), which reflects a baseline difficulty shift rather than a mapping difference.

Because MMLU items are clustered within 57 subjects, HC3 covariances underestimate standard errors (Liang and Zeger, 1986). The Bell-McCaffrey CR2 correction (Bell and McCaffrey, 2002) inflates slope standard errors by a median factor of 1.65, reducing the joint statistic to $W(9) = 28.21$ ($p = 8.8 \times 10^{-4}$; $W(9) = 33.13$, $p = 1.3 \times 10^{-4}$ under CR1), which still rejects against χ^2_9 , and because 57 is a small number of clusters we further apply a wild cluster bootstrap with the null imposed (Cameron et al., 2008), which yields an anti-conservative reference distribution (95th percentile 34.7 against nominal 16.9) and renders the joint test marginal ($p = 0.060$). Since a 9-degree-of-freedom test dilutes power across contrasts already shown to be null, bootstrapping one contrast at a time proves decisive for Entity Density ($p_{\text{Holm}} = 0.014$) and suggestive for Syntactic MDD ($p = 0.015$ uncorrected). We report the nested likelihood ratio test ($\Lambda(11) = 454.86$) for comparability with the invariance literature (Kline, 2016; Wilks, 1938) but rely on the resampling tests, since the residuals exhibit significant skew (1.37) and heteroskedasticity (Breusch-Pagan 130.70).

Permutation tests confirm the necessity of cluster-aware inference. An item-level permutation of the STEM label yields a maximum R^2 gap of 0.0315, below the observed 0.0412 ($p < 10^{-4}$), but that null ignores between-subject heterogeneity: permuting whole 19-subject blocks widens it by a factor of 3.2 and renders the gap non-

significant ($p = 0.08$), while the slope-sensitive likelihood-ratio statistic still clears its block null ($p = 0.049$). Whether structural complexity predicts difficulty *better* in STEM is therefore sensitive to cluster structure, but whether it predicts difficulty *differently* is robust. Independent of these standard errors, the confirmatory two-dimensional IRT model (Appendix C) yields a latent correlation of $\rho = 0.965$ with no posterior mass above 0.99, improves held-out prediction by +243.6 nats on 5% masked response cells (paired $z = 8.2$), and is corroborated by a model-free matched-length subscale correlation of 0.977 over 200 random splits ($[0.974, 0.980]$), with no replicate reaching unity.

Structural homogeneity is therefore rejected: the mapping from text-level complexity to IRT-estimated difficulty is not constant across partitions, an item-side analogue to measurement non-equivalence (Meredith, 1993). This licenses claims about *which* structural properties carry difficulty in each domain, not that one domain affords more predictable difficulty overall. The conclusion survives a Rasch refit (Appendix A), holds on each predictor’s common support controlling for option-set indicators (Appendix B), and subsists after removing every item flagged as erroneous by MMLU-Redux (Appendix D).

5 Consequences for Leaderboard Validity

If the complexity-to-difficulty mapping differs structurally across domains, the interpretation of the aggregate as a unified ability measure is undermined. Because 77.55% of the items are non-STEM, the aggregate is arithmetically weighted toward non-STEM performance to begin with, and the non-invariance established in Section 4.3 is what makes that weighting a measurement asym-

metry rather than a counting artifact. Two consequences follow.

5.1 Reasoning Stability

A Reasoning Sensitivity Coefficient (β_{sj}) can be estimated per model by logistic regression of binary accuracy on WSCG Depth. Restricted to STEM and fitted in log-odds space, β_{sj} correlates with latent ability at $r = -0.772$ across all 1,000 models, and at -0.512 and -0.397 once the bottom quartile and bottom half are excluded, suggesting that higher-ability models degrade more steeply as reasoning depth increases. This two-stage estimate is, however, severely confounded by the 25% guessing floor: because success probability is bounded at $c = 0.25$, empirical log-odds are mechanically compressed for weaker models, flattening their slopes relative to stronger ones even absent any true ability-by-complexity interaction. Simulating response matrices from a 3PL model ($c = 0.25$) fitted to the STEM submatrix and passing them through the identical procedure isolates that geometric artifact: the generative null yields correlations of -0.899 , -0.355 and -0.119 across the three samples, accounting for 116%, 69% and 30% of the observed values while preserving construct consistency (Figure 1), its abilities and difficulties correlating at 0.907 and 0.892 with the 2PL baseline.

To handle the guessing floor natively and avoid interpreting two-stage residuals, we replace this procedure by estimating the interaction directly inside the response model,

$$\eta_{ji} = a_i(\theta_j - b_i) + (\gamma_0 + \gamma_1\theta_j)W_i,$$

with $P(Y_{ji} = 1) = c + (1 - c)\sigma(\eta_{ji})$, $c = 0.25$ and W_i standardized, so that γ_1 captures the interaction as a native parameter. Structural complexity depresses success at mean ability ($\gamma_0 = -0.300$, CI $[-0.326, -0.272]$), and the interaction remains negative but highly modest, at $\gamma_1 = -0.086$ logits per standard deviation of WSCG Depth per unit of ability (CI $[-0.093, -0.079]$). Higher-ability models *do* therefore degrade more steeply under increased reasoning depth, but the true effect is much smaller than the naive correlation implies. Two qualifications apply: mean-field intervals understate posterior spread, so the point estimate’s sign and magnitude are the takeaway, and the effect is estimated strictly on the STEM partition.

5.2 Consequences for Downstream Model Selection

In construct validity terms (Cronbach and Meehl, 1955; Messick, 1989), the MMLU aggregate conflates Retrieval Capacity and Reasoning Stability, weighting the former more heavily, and practically this affects rank stability. Weighted Kendall’s τ (Shieh, 1998; Vigna, 2015), which prioritizes frontier rank agreement, is higher for non-STEM ($\tau_w = 0.9887$) than for STEM (0.9634); subsampling non-STEM to STEM’s 3,153 items over 200 draws halves the gap to a matched 0.9756 (CI $[0.9633, 0.9827]$), and the STEM value sits below 194 of these 200 draws ($p = 0.035$), confirming a real, though reduced, one-sided asymmetry.

The asymmetry manifests directly in model selection. Of the Top 50 aggregate models, 92% retain their status on non-STEM but only 78% do so on STEM, and for the Top 10 retention is 100% against 80%, so relying on the aggregate leaderboard for a STEM-focused application carries a 22% displacement rate (bootstrap CI $[16\%, 28\%]$). This displacement is largely driven by the smaller STEM item sample, since a size-matched random-item null yields 14.4% ($p = 0.086$) and a subject-block matched null 22.4% ($p = 0.61$); although deduplicating to the 869 base models raises it to 28.8%, we report the metric as a quantified cost to practitioners rather than as independent evidence for construct separation. As an empirical noise floor (Alzahrani et al., 2024), seven duplicate checkpoint pairs agreeing within 0.1% aggregate accuracy disagree by a median of 1 aggregate and 3 STEM ranks (maxima 11 and 23), so shifts below roughly ten places are largely noise. Evaluation noise does not, however, displace models in one direction: standardizing both abilities of the two-dimensional model, the $\theta_{\text{STEM}} - \theta_{\text{non-STEM}}$ gap has a standard deviation of 0.19, and while 83.5% of models sit within 0.25σ of the identity line, the positive tail is systematic and belongs to reasoning-tuned models such as Qwen2-72B (+0.88). The aggregate summarizes the general population adequately while being unrepresentative of a minority optimized for procedural reasoning.

6 Discussion

6.1 Implications for Benchmark Design

The heuristic reforms of MMLU-Pro (Wang et al., 2024) are useful but theoretically insufficient, since filtering by difficulty or discrimination removes

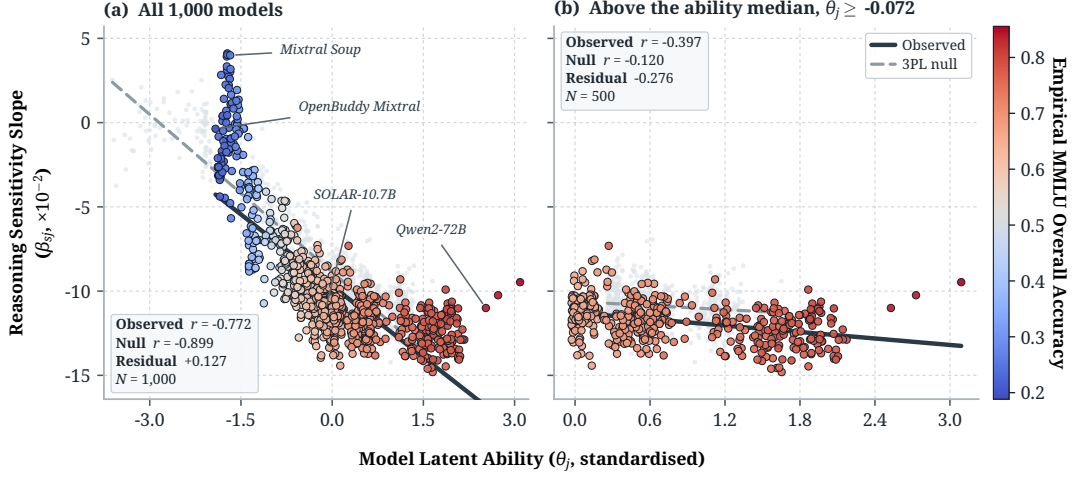


Figure 1: Reasoning sensitivity slopes (β_{sj}) against latent ability (θ_j): (a) all 1,000 models, (b) the upper-ability half. Filled/solid: observed. Open/dashed: 3PL nulls without ability-by-complexity interaction.

uninformative items and leaves the non-invariance of the mapping untouched, whereas our deterministic framework supplies the missing structural audit without the stochastic variance and API-dependence of LLM-as-annotator methods (Gilardi et al., 2023). We recommend that future benchmarks report distinct retrieval and reasoning capacity indices alongside aggregate scores. Unlike category subscores, these are computable *a priori* from item text before any model is evaluated, are explanatory in distinguishing whether a domain is difficult because it demands deep inference or because it presupposes rare facts, and are independent of designer taxonomies, which routinely misalign with structural reality: MMLU’s *Business Ethics* and *High School Macroeconomics* are highly structurally sensitive, yet labeled non-STEM. Applying the framework to MMLU-Pro will test whether its construction resolves structural heterogeneity or merely displaces it upward.

6.2 Limitations

Variance explained and scope. As the dense encoder ceiling establishes (Section 4.1), structural complexity is a significant but not dominant determinant of difficulty. Our instrument reads text only, and because MMLU’s mathematical content is overwhelmingly ASCII, the top symbolic weight tier is inert and Zipf rarity defaults to the population mean for items lacking scorable content words, so the instrument is weakest exactly where the abstract STEM signal is strongest. This evaluation also targets open-weights models, and whether these sensitivities extend to closed frontier models remains open.

Statistical and methodological bounds. Several findings are marginal against their strictest respective controls: the between-domain R^2 gap against a subject-block permutation ($p = 0.08$), the Top-50 displacement rate against a size-matched control ($p = 0.09$), and the joint homogeneity test under a wild cluster bootstrap ($p = 0.060$). We adjust for effective sample size by modeling 57 subject clusters rather than 14,042 items and 50 base-model families rather than 1,000 checkpoints, which renders individual subject-level inferences underpowered, and the confounds we control explicitly, namely prompt length, answer-option structure, label error and pretraining contamination (Appendices B and D), are bounded rather than eliminated. The ideal specification would be a fully explanatory IRT model propagating difficulty uncertainty into the domain contrast while natively absorbing subject clustering; we were unable to make that model identify, and Appendix C documents the attempt as a formal negative result.

Construct invariance limits. The rejection of structural homogeneity is an *item-side* phenomenon and does not formally license *person-side* measurement non-equivalence (Meredith, 1993); our design does not strictly prove a single latent θ incomparable across models. Mantel–Haenszel tests do reveal differential item functioning (Appendix D), but that is a supplementary finding rather than a corollary of the structural mapping shift.

7 Conclusion

IRT calibration, multi-group analysis and confirmatory two-dimensional IRT applied to 14 million response observations from 1,000 language mod-

els demonstrate that MMLU’s partitions do not measure a single interchangeable ability: the latent correlation between STEM and non-STEM ability is strictly below unity at $\rho = 0.965$, the disattenuated subscale correlation at matched test length is 0.977, and the two-dimensional model improves held-out prediction. Crucially, the mapping from text structure to item difficulty is structurally non-invariant across partitions, a rejection holding under item-level inference ($W(9) = 138.36$) and subject-clustered covariances ($W(9) = 28.21$ under CR2), and the divergence is most absolute for named entity density, which acts as an operational constraint compounding computational load in STEM items but as an associative retrieval cue elsewhere. The heterogeneity carries downstream consequences: Top-50 aggregate selection displaces 22% of STEM-appropriate choices, and once the guessing floor is controlled natively, higher-ability models degrade more steeply under procedural reasoning depth ($\gamma_1 = -0.086$). Because the MMLU aggregate weights declarative retrieval capacity over procedural reasoning stability, it inherently favors models optimized for retrieval. We release the framework so that future benchmarks can be structurally audited before they are adopted as unified capability measures.

References

- Norah Alzahrani, Hisham Abdullah Alyahya, Yazeed Alnumay, Sultan AlRashed, Shaykhah Alsubaie, Yousef Almushayqih, Faisal Mirza, Nouf Alotaibi, Nora Al-Twairsh, Areeb Alowisheq, M. Saiful Bari, and Haidar Khan. 2024. When benchmarks are targets: Revealing the sensitivity of large language model leaderboards. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13787–13805.
- John R. Anderson. 1983. *The Architecture of Cognition*. Harvard University Press, Cambridge, MA.
- Michael A. Babyak. 2004. What you see may not be what you get: A brief, nontechnical introduction to overfitting in regression-type models. *Psychosomatic Medicine*, 66(3):411–421.
- Laura Banarescu, Claire Bonial, Shu Cai, Madalina Georgescu, Kira Griffitt, Ulf Hermjakob, Kevin Knight, Philipp Koehn, Martha Palmer, and Nathan Schneider. 2013. Abstract meaning representation for sembanking. In *Proceedings of the 7th Linguistic Annotation Workshop and Interoperability with Discourse*, pages 178–186.
- Edward Beeching, Clémentine Fourier, Nathan Habib, Sheon Han, Nathan Lambert, Nazneen Rajani, Omar Sanseviero, Lewis Tunstall, and Thomas Wolf. 2023. [Open LLM leaderboard](#). Hugging Face.
- Robert M. Bell and Daniel F. McCaffrey. 2002. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28(2):169–181.
- Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FACCT ’21)*, pages 610–623, New York, NY, USA. Association for Computing Machinery.
- Rita Sloan Berndt, Charlotte C. Mitchum, and Anne N. Haendiges. 1996. Comprehension of reversible sentences in “agrammatism”: A meta-analysis. *Cognition*, 58(3):289–308.
- Allan Birnbaum. 1968. Some latent trait models and their use in inferring an examinee’s ability. In Frederic M. Lord and Melvin R. Novick, editors, *Statistical Theories of Mental Test Scores*, pages 397–479. Addison-Wesley, Reading, MA.
- Samuel R. Bowman and George Dahl. 2021. What will it take to fix benchmarking in natural language understanding? In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4843–4855.
- Trevor S. Breusch and Adrian R. Pagan. 1979. A simple test for heteroscedasticity and random coefficient variation. *Econometrica*, 47(5):1287–1294.
- Marc Brysbaert, Amy Beth Warriner, and Victor Kuperman. 2014. Concreteness ratings for 40 thousand generally known English word lemmas. *Behavior Research Methods*, 46(3):904–911.

- A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. 2008. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427.
- John B. Carroll. 1993. *Human Cognitive Abilities: A Survey of Factor-Analytic Studies*. Cambridge University Press, Cambridge, UK.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *ArXiv:1803.05457*.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.
- Jacob Cohen. 1988. *Statistical Power Analysis for the Behavioral Sciences*, 2nd edition. Lawrence Erlbaum Associates, Hillsdale, NJ.
- Lee J. Cronbach and Paul E. Meehl. 1955. Construct validity in psychological tests. *Psychological Bulletin*, 52(4):281–302.
- Paul De Boeck and Mark Wilson. 2004. *Explanatory Item Response Models: A Generalized Linear and Nonlinear Approach*. Springer, New York.
- Vera Demberg and Frank Keller. 2008. Data from eye-tracking corpora as evidence for theories of syntactic processing complexity. *Cognition*, 109(2):193–210.
- Chunyuan Deng, Yilun Zhao, Xiangru Tang, Mark Gerstein, and Arman Cohan. 2024. Investigating data contamination in modern benchmarks for large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 8706–8719.
- Neil J. Dorans and Paul W. Holland. 1993. DIF detection and description: Mantel–Haenszel and standardization. In Paul W. Holland and Howard Wainer, editors, *Differential Item Functioning*, pages 35–66. Lawrence Erlbaum Associates.
- Susan E. Embretson. 1983. Construct validity: Construct representation versus nomothetic span. *Psychological Bulletin*, 93(1):179–197.
- Susan E. Embretson and C. Douglas Wetzel. 1987. Component latent trait models for paragraph comprehension tests. *Applied Psychological Measurement*, 11(2):175–193.
- Gerhard H. Fischer. 1973. The linear logistic test model as an instrument in educational research. *Acta Psychologica*, 37(6):359–374.
- Aryo Pradipta Gema, Joshua Ong Jun Leang, Giwon Hong, Alessio Devoto, Alberto Carlo Maria Mancino, Rohit Saxena, Xuanli He, Yu Zhao, Xiaotang Du, Mohammad Reza Ghasemi Madani, Claire Barale, Robert McHardy, Joshua Harris, Jean Kaddour, Emile van Krieken, and Pasquale Minervini. 2025. Are we done with MMLU? In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5069–5096.
- Edward Gibson. 1998. Linguistic complexity: Locality of syntactic dependencies. *Cognition*, 68(1):1–76.
- Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. 2023. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120.
- Daniel Gildea and T. Florian Jaeger. 2015. Human languages order information efficiently. *ArXiv:1510.02823*.
- Shahriar Golchin and Mihai Surdeanu. 2024. Time travel in LLMs: Tracing data contamination in large language models. In *The Twelfth International Conference on Learning Representations*.
- Frank E. Harrell. 2015. *Regression Modeling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis*, 2nd edition. Springer.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021a. Measuring massive multi-task language understanding. In *Proceedings of the 9th International Conference on Learning Representations (ICLR 2021)*.

- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. 2021b. Measuring mathematical problem solving with the MATH dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*.
- Paul W. Holland and Dorothy T. Thayer. 1988. Differential item performance and the Mantel-Haenszel procedure. In Howard Wainer and Henry I. Braun, editors, *Test Validity*, pages 129–145. Lawrence Erlbaum Associates.
- Rex B. Kline. 2016. *Principles and Practice of Structural Equation Modeling*, 4th edition. Guilford Publications, New York, NY.
- John P. Lalor, Hao Wu, and Hong Yu. 2019. Learning latent parameters without human response patterns: Item response theory with artificial crowds. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 4249–4259, Hong Kong, China.
- Roger Levy. 2008. Expectation-based syntactic comprehension. *Cognition*, 106(3):1126–1177.
- Kung-Yee Liang and Scott L. Zeger. 1986. Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1):13–22.
- Haitao Liu. 2008. Dependency distance as a metric of language comprehension difficulty. *Journal of Cognitive Science*, 9(2):159–191.
- Frederic M. Lord and Melvin R. Novick. 1968. *Statistical Theories of Mental Test Scores*. Addison-Wesley, Reading, MA.
- James G. MacKinnon and Halbert White. 1985. Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3):305–325.
- R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3428–3448, Florence, Italy.
- William Meredith. 1993. Measurement invariance, factor analysis and factorial invariance. *Psychometrika*, 58(4):525–543.
- Samuel Messick. 1989. Validity. In Robert L. Linn, editor, *Educational Measurement*, 3rd edition, pages 13–103. American Council on Education/Macmillan, New York, NY.
- Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. 2024. State of what art? a call for multi-prompt LLM evaluation. *Transactions of the Association for Computational Linguistics*, 12:933–949.
- Nora S. Newcombe, Nalini Ambady, Jacquelynne Eccles, Louis Gomez, David Klahr, Marcia Linn, Kevin Miller, and Kelly Mix. 2009. Psychology’s role in mathematics and science education. *American Psychologist*, 64(6):538–550.
- Jerzy Neyman and Elizabeth L. Scott. 1948. Consistent estimates based on partially consistent observations. *Econometrica*, 16(1):1–32.
- Pouya Pezeshkpour and Estevam Hruschka. 2024. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2006–2017.
- Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. 2024. tinyBenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*.
- Inioluwa Deborah Raji, Emily M. Bender, Amanda-lynn Paullada, Emily Denton, and Alex Hanna. 2021. AI and the everything in the whole wide world benchmark. In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS 2021), Track on Datasets and Benchmarks*.
- Georg Rasch. 1960. Probabilistic models for some intelligence and attainment tests. *Danish Institute for Educational Research, Copenhagen*.
- Mark D. Reckase. 2009. *Multidimensional Item Response Theory*. Springer, New York, NY.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese

- BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 3982–3992.
- Pedro Rodriguez, Joe Barrow, Alexander Miserlis Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. 2021. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4486–4503.
- Anna Rogers, Olga Kovaleva, and Anna Rumshisky. 2020. A primer in BERTology: What we know about how BERT works. *Transactions of the Association for Computational Linguistics*, 8:842–866.
- Oscar Sainz, Jon Ander Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10776–10787.
- Grace S. Shieh. 1998. A weighted Kendall’s tau statistic. *Statistics & Probability Letters*, 39(1):17–24.
- Charles Spearman. 1904. “general intelligence,” objectively determined and measured. *The American Journal of Psychology*, 15(2):201–292.
- Robyn Speer. 2022. [wordfreq: An open-source package for word frequencies in 36 languages](#). Zenodo. Version 3.0.2.
- Mervyn Stone. 1974. Cross-validatory choice and assessment of statistical predictions. *Journal of the Royal Statistical Society, Series B*, 36(2):111–147.
- David Temperley. 2007. Minimization of dependency length in written English. *Cognition*, 105(2):300–333.
- Louis L. Thurstone. 1938. *Primary Mental Abilities*. Psychometric Monographs, No. 1. University of Chicago Press, Chicago, IL.
- Endel Tulving. 1972. Episodic and semantic memory. In Endel Tulving and Wayne Donaldson, editors, *Organization of Memory*, pages 381–403. Academic Press, New York, NY.
- Robert J. Vandenberg and Charles E. Lance. 2000. A review and synthesis of the measurement invariance literature: Suggestions, practices, and recommendations for organizational research. *Organizational Research Methods*, 3(1):4–70.
- Clara Vania, Phu Mon Htut, William Huang, Dhara Mungra, Richard Yuanzhe Pang, Jason Phang, Haokun Liu, Kyunghyun Cho, and Samuel R. Bowman. 2021. Comparing test sets with item response theory. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1141–1158.
- Sebastiano Vigna. 2015. A weighted correlation index for rankings with ties. In *Proceedings of the 24th International Conference on World Wide Web (WWW)*, pages 1166–1176. Association for Computing Machinery.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuan-sheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Proceedings of the 38th Annual Conference on Neural Information Processing Systems (NeurIPS 2024), Track on Datasets and Benchmarks*.
- Halbert White. 1980. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4):817–838.
- Samuel S. Wilks. 1938. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62.
- Tal Yarkoni and Jacob Westfall. 2017. Choosing prediction over explanation in psychology: Lessons from machine learning. *Perspectives on Psychological Science*, 12(6):1100–1122.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4791–4800.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. 2024. Large language models are not robust multiple choice selectors. In *The Twelfth International Conference on Learning Representations*.

Wanjun Zhong, Ruixiang Cui, Yiduo Guo, Yaobo Liang, Shuai Lu, Yanlin Wang, Amin Saied, Weizhu Chen, and Nan Duan. 2024. AGIEval: A human-centric benchmark for evaluating foundation models. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2299–2314.

A Sensitivity, Diagnostics and Clustering

Weight sensitivity. We apply 123 weight tier perturbation schemes clamped to $[0.0, 5.0]$: uniform shifts, single-tier shifts and 100 Monte Carlo draws from $[-1, +1]$. All primary conclusions hold across all 123 schemes, with structural homogeneity universally rejected (maximum $p = 1.26 \times 10^{-77}$), STEM R^2 strictly dominating non-STEM R^2 (0.077–0.083 against 0.031–0.043) and the Syntactic MDD divergence remaining positive ($p < 0.01$, $Z \in [3.22, 4.54]$).

Diagnostics and robustness. Partition-level fits are stable under 20 iterations of 5-fold cross-validation (Stone, 1974), at R_{cv}^2 of 0.0732 and 0.0361 against in-sample 0.0792 and 0.0380, though individual subject pools show the standard small- N failure mode (Babyak, 2004; Yarkoni and Westfall, 2017). Residuals are right-skewed (1.37, 0.89, 1.53), Jarque–Bera rejects normality and Breusch–Pagan (Breusch and Pagan, 1979) and White (White, 1980) tests reject homoskedasticity (LM 130.70, 42.05, 120.84), which necessitates resampling rather than normal-theory inference. Refitting difficulty via a Rasch model (Rasch, 1960) recovers estimates correlating at $r = 0.894$ with the 2PL baseline and strengthens the contrast (0.1038 against 0.0676, $W(9) = 167.21$); filtering uninformative items ($a_i < 0.2$), which disproportionately affects STEM (31.9% of removals against a 22.5% pool share), raises STEM R^2 to 0.0862 while leaving non-STEM invariant

($W(9) = 149.58$), and slope equality is rejected at every tested threshold.

Clustering and size-matched nulls. Predictor variances differ across partitions (WSCG Nodes has a $2.82\times$ larger standard deviation outside STEM), but standardizing within partitions leaves the identical five contrasts clearing Holm correction without sign reversals. Permuting whole 19-subject blocks rather than items widens the R^2 -gap null more than threefold, to mean 0.0076 (max 0.1056, $p = 0.080$), and the block null for rank displacement at $K = 50$ is 22.4% ($p = 0.609$) against a random-item null of 14.4% ($p = 0.086$). Evaluated one contrast at a time under the wild cluster bootstrap, only Entity Density survives Holm correction ($p = 0.0015$ raw, 0.014 corrected), with Syntactic MDD next at 0.015 raw and 0.12 corrected.

B Functional Form and Option Structure

Common support and functional form. Restricting partitions to their intersected 1st–99th percentile ranges retains 97–99% of items and preserves all conclusions: Syntactic MDD shifts to 0.4083/0.0168 ($Z = 3.90$), Entity Density to 0.0353/0.0036 ($Z = 5.49$) and Concreteness to $-0.1479/0.7601$ ($Z = -5.13$). A four-knot restricted cubic spline (Harrell, 2015) on the focal indicator significantly improves fit outside STEM for Syntactic MDD ($F(2, 10877) = 17.43$) and Concreteness ($F = 15.86$) but not inside it ($p > 0.15$): Syntactic MDD and Entity Density rise monotonically in STEM but trace non-monotone curves outside it.

Option-set indicators. We extract eight choice-based indicators, for instance distractor homogeneity, option length dispersion and numeric proximity, using a frozen MiniLM sentence encoder (Reimers and Gurevych, 2019). These reach $R_{cv}^2 = 0.0684$ globally against 0.0338 for the stem indicators, and the two families combined reach 0.0808 globally and 0.1196 in STEM. Refitting the homogeneity test on the 17-indicator set remains decisive ($W(17) = 138.86$), and the option indicators are themselves non-invariant ($W(8) = 51.97$). Their two strongest members are the key’s stem-similarity advantage ($b = -1.477$) and maximum distractor similarity ($b = 1.056$), so difficulty scales in direct proportion to how closely an item’s best distractor imitates its key.

C Confirmatory Two-Dimensional IRT

Each model receives a STEM ability and a non-STEM ability drawn from a bivariate normal with a free correlation, each item loads only on the dimension its designer-fixed subject assigns, and the model is estimated by SVI as in Section 3.1. The latent correlation is 0.965 with no posterior mass above 0.99, and the variational objective improves over the unidimensional model. Because an improvement in a bound on in-sample fit is weak evidence when the larger model carries a thousand additional parameters, we also mask a random 5% of response cells before estimation and score each model on cells it never observed: the two-dimensional model is better there by +243.6 nats, with a paired z of 8.2 over 703,696 held-out cells. Disattenuating matched-length subscale scores by their Spearman–Brown reliabilities gives 0.977 over 200 random splits at 1,576 items per side ($[0.974, 0.980]$), and because mean-field posteriors understate spread we rely on that split-half estimate for uncertainty.

We also fitted the natural companion analysis, an explanatory IRT model writing difficulty as a function of the indicators inside the response model with domain-varying slopes and a subject random effect (Fischer, 1973; De Boeck and Wilson, 2004), but do not report its estimates, because they are not trustworthy: the fitted residual item scale (4.93 logits) exceeds the standard deviation of the difficulty distribution itself (2.24), the credible intervals are an order of magnitude narrower than the corresponding robust standard errors, and refitting at increasing step budgets reverses the sign of two of the nine domain interactions. The model is weakly identified as written, since a free per-item residual over 14,042 items absorbs what the indicators would otherwise explain.

D Label Errors, Contamination and Transfer

Label error and contamination. Matching the Gema et al. (2025) re-annotation to our pool covers 5,984 items and flags 374 (6.25%), at a lower rate in STEM (4.70%) than outside (6.98%). As expected for incorrect keys, flagged items are significantly harder (mean $b = +1.86$ against -0.10 , $t = 12.08$), but they do not drive the domain contrast: dropping them yields STEM and non-STEM R^2 of 0.0852 and 0.0442 ($W(9) = 140.71$), refitting on the verified subset alone yields 0.0934

and 0.0141 ($W(9) = 84.45$), and leaving the annotated subsample unfiltered yields 0.0828 and 0.0115 ($W(9) = 82.79$), so the divergence is intrinsic to the subsample rather than an artifact of filtering. Recalibrating difficulty on an early cohort (99 models: Pythia, Falcon, Llama-2) and a recent one (606 models: Llama-3, Qwen2, Phi-3, Gemma, Mixtral, Yi-1.5) yields difficulty vectors correlated at $r = 0.832$, 86% of the attainable $r = 0.967$ ceiling, and the contrast persists in both, at 0.0712/0.0505 ($W(9) = 124.14$) and 0.0839/0.0577 ($W(9) = 135.81$); under the strictest control, flagged items removed and difficulty estimated on the early cohort alone, fits are 0.0751 and 0.0552.

Respondent-side invariance and transfer. Mantel–Haenszel DIF (Holland and Thayer, 1988; Dorans and Holland, 1993), stratifying models into ability deciles with Llama ($n = 319$) as reference, classifies 42.9% of items as B or C for Mixtral, 44.9% for Solar and 30.2% for Yi, and the proportion flagged is higher in STEM across all three families by 7.5, 7.5 and 7.2 points (all $z > 7.4$), an excess reproduced by a purified second pass over the 4,714 items classified A throughout. Transferring the stem-only framework to HellaSwag (Zellers et al., 2019) returns a clean null across the designer-fixed ActivityNet/WikiHow boundary ($W(8) = 3.64$, $p = 0.89$), whereas the MMLU control refitted on the identical 219 models reproduces the bifurcation ($W(8) = 150.48$) and six ending-set indicators reject source invariance on those same HellaSwag items ($W(6) = 65.33$), so the test does not mechanically fire on arbitrary large partitions. On GSM8K (Cobbe et al., 2021) the WSCG tracks annotated calculator-step counts beyond prompt length (partial $r = 0.155$, $p = 1.9 \times 10^{-8}$), whereas on MATH (Hendrycks et al., 2021b) it fails to track human-assigned difficulty once length is controlled. Deduplicated re-runs across the 869 checkpoints reproduce the ability-to-sensitivity correlations to within 0.002, so redundancy inflates significance rather than effect size.